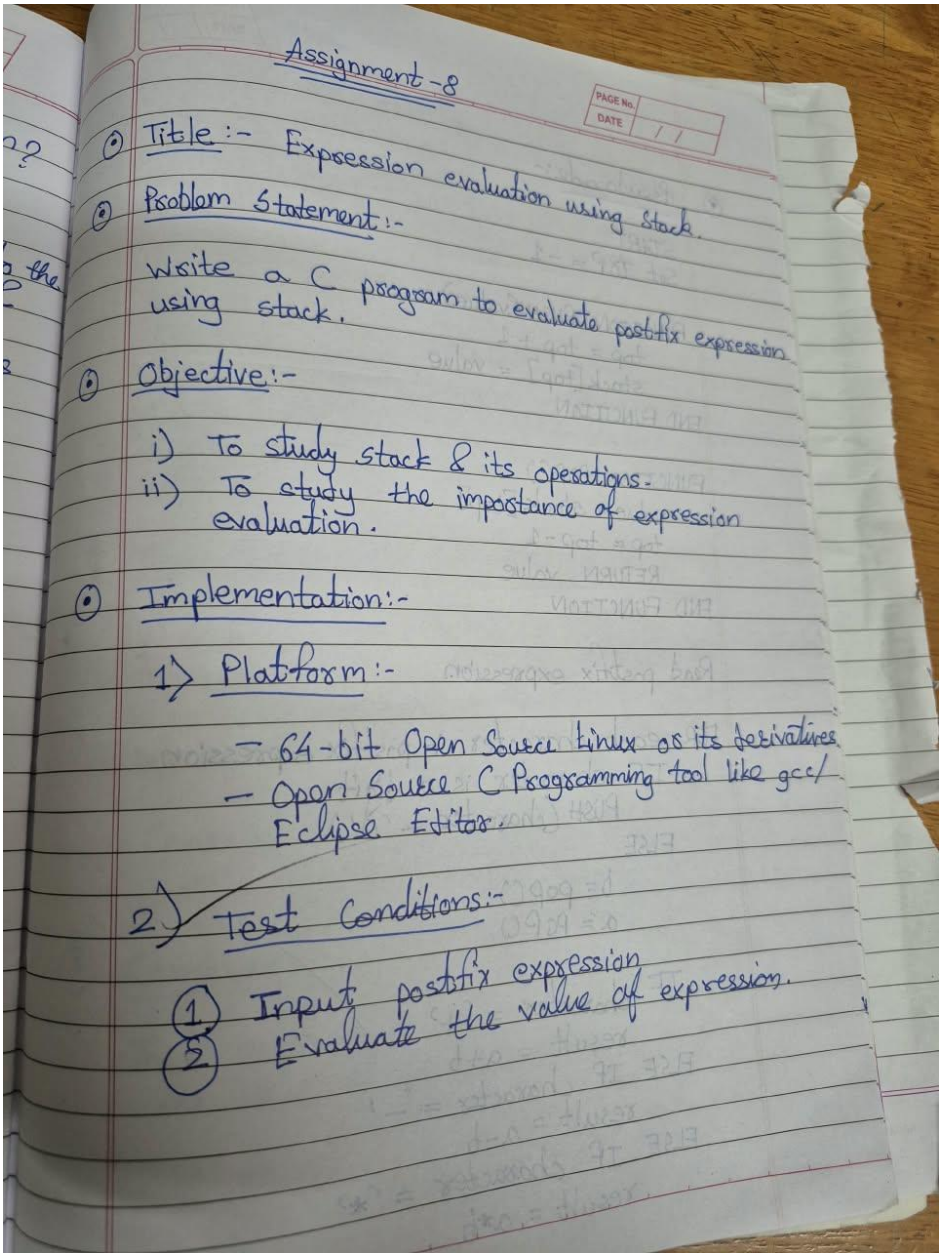
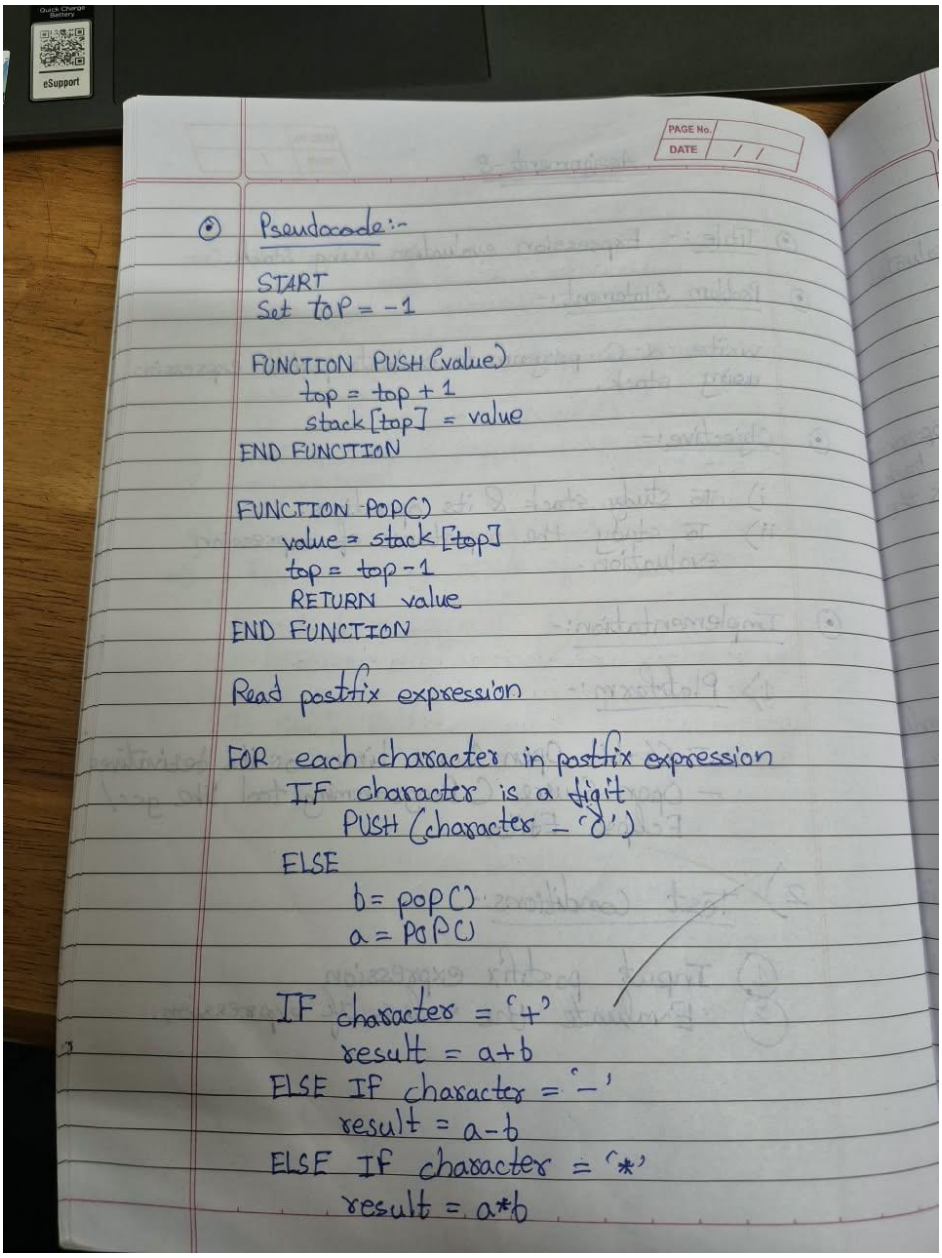
ASSIGNMENT-8

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BATCH - A1





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⊙ Pseudocode:-

```
START
Set top = -1

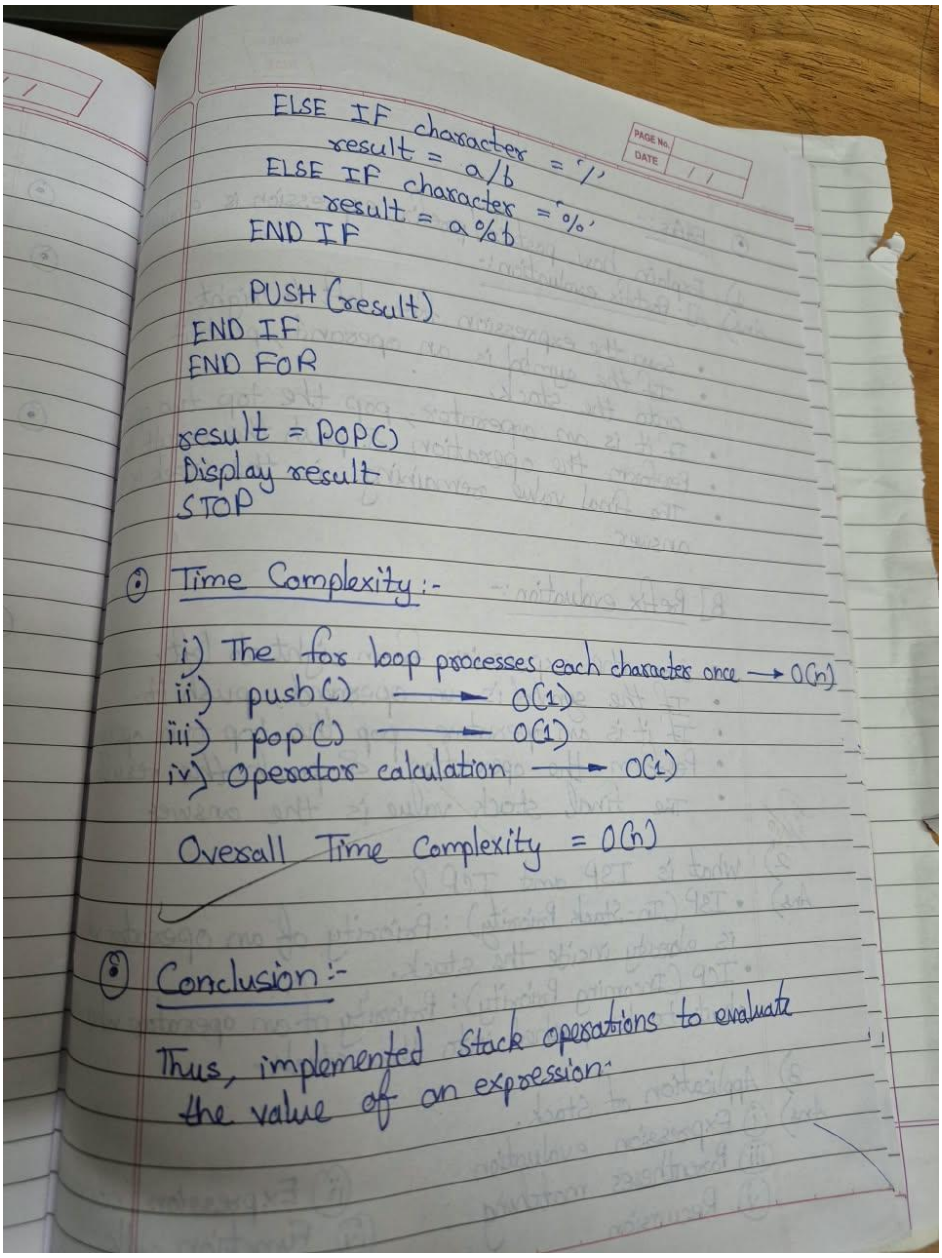
FUNCTION PUSH (value)
    top = top + 1
    stack[top] = value
END FUNCTION

FUNCTION POP()
    value = stack[top]
    top = top - 1
    RETURN value
END FUNCTION

Read postfix expression

FOR each character in postfix expression
    IF character is a digit
        PUSH (character - '0')
    ELSE
        b = pop()
        a = POP()

    IF character = '+'
        result = a+b
    ELSE IF character = '-'
        result = a-b
    ELSE IF character = '*'
        result = a*b
```



```
        ELSE IF character = '/'
            result = a/b
        ELSE IF character = '%'
            result = a%b
        END IF

        PUSH (result)
    END IF
END FOR

result = POP()
Display result
STOP
```

⑦ Time Complexity :-

i) The for loop processes each character once → O(n)
ii) push() → O(1)
iii) pop() → O(1)
iv) Operator calculation → O(1)

Overall Time Complexity = O(n)

⑧ Conclusion :-

Thus, implemented Stack operations to evaluate the value of an expression.

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⊙ FAQs :-

1) Explain how postfix / prefix expression is evaluated.

Ans) A] Postfix evaluation :-

- Scan the expression from left to right.
- If the symbol is an operand, push it onto the stack.
- If it is an operator, pop the top two operands.
- Perform the operation & push the result back.
- The final value remaining in the stack is the answer.

B] Prefix evaluation :-

- Scan the expression from right to left.
- If the symbol is an operand, push it.
- If it is an operator, pop the top two operands.
- Perform the operation & push the result.
- The final stack value is the answer.

2) What is ISP and ICP ?

Ans)
- ISP (In-Stack Priority) : Priority of an operator when it is already inside the stack.
- ICP (Incoming Priority) : Priority of an operator when it is about to be pushed into the stack.

3) Application of Stack.

Ans)
(i) Expression evaluation
(ii) Expression conversion
(iii) Parentheses matching
(iv) Function calls
(v) Recursion
(vi) Undo/Redo operations